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## Education

University of California, San Diego
B.S. Electrical Engineering - Computer System Design Depth
**Relevant Coursework:** Rapid Hardware and Software Design(C++ & Python), Software Foundations(C++), Data Structures in C++, Programming for Data Analysis(Python), Embedded Systems, Data Networks, Power System Analysis

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## Skills

- **Programming Languages:** C++, Python, SQL, C, MATLAB, SystemVerilog
- **Low-level Programming:** Manual memory management, resource allocation, modular embedded firmware design
- **Interfaces:** ADC, GPIO, SPI, I2C
- **Tools & Frameworks:** Git, FastAPI, MySQL, NumPy, Matplotlib, Pandas, Arduino
- **Troubleshooting & Debugging:** Embedded Systems, Software, Serial Communication, Signal Integrity, Multimeter/Oscilloscope use
- **Soft Skills:** Problem-solving, fast learner, collaborative, detail oriented

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## Projects - selected projects demonstrating embedded systems development, real-time communication, low-level programming, and cross-layer debugging

**Smartwatch Prototype (Embedded Systems, Arduino, C++, Python)**

- Designed and implemented an IoT device with **90%+ pedometer accuracy, real-time heart rate monitoring, idle detection, and weather updates**
- Developed a **Python-based data processing pipeline** that used **GMM-based peak detection** to estimate heart rate and step count in real-time
- Built hardware using microcontroller, photodetector, accelerometer, and OLED
- Interfaced sensors and display using **I2C/SPI** and debugged real-time firmware for accuracy & stability

**Decentralized Communication Device (Embedded Systems, C++, ESP-NOW)**

- Designed and built a portable, wireless messaging device for use in remote, no-signal environments using ESP32 microcontrollers
- Implemented message parsing, inbox storage, and multi-device mesh-like communication logic
- Utilized **GPIO** button handling; implemented memory-efficient message relay system
- Refactored codebase into modular components (communication stack, **display driver**, input logic) for scalability and maintainability
- **Collaborated** with team to debug transmission issues and field-test performance in real-world conditions

**C++ Software Development Projects**

- **Buffer Manager & Custom String Class -** Designed an efficient **memory management system** for my own **custom C++ string class**, optimizing dynamic memory allocation and reducing redundant copies
- **Custom Variant Class** - Implemented a **type-safe class** for handling multiple data types without relying STL variant class
- **Archive System** - Designed and developed a **block-based file archiving system** using **chunked data storage** for optimal file management

**Restaurant Ordering Web App (Fullstack: HTML, CSS, Javascript, FastAPI, SQL)**

- Developed a restaurant ordering web app supporting user and employee roles, with **user registration, login,** and **session-based authentication** using **FastAPI** and **MySQL**
- Created a separate worker dashboard to display incoming orders and update order status